

Fenoprazil Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-146/A

Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

No changes to the approved analytical procedures are proposed in this submission. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control.

Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Deviations observed during the campaign were investigated and closed with no impact to product quality. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Cleaning verification swab results were below the health-based exposure limits derived for the product. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

Facilities and Equipment

Manufacturing is performed at the approved commercial site using dedicated product-contact equipment trains. Equipment qualification and cleaning validation are maintained under the site validation master plan. No changes to the approved facility registration are proposed.

Statistical evaluation of batch release data demonstrates process capability well within specification limits. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.

Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. No changes to the approved analytical procedures are proposed in this submission. Cleaning verification swab results were below the health-based exposure limits derived for the product. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions.

Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Statistical evaluation of batch release data demonstrates process capability well within specification limits. The control strategy links critical quality attributes to critical process parameters identified during development. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results.

Analytical Method Validation Summary

The assay and related-substances procedures employ reversed-phase high performance liquid chromatography with ultraviolet detection. Validation demonstrated acceptable specificity, linearity, accuracy, precision, and robustness. Forced degradation studies confirmed the stability-indicating capability of the related-substances method.

The control strategy links critical quality attributes to critical process parameters identified during development. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. No changes to the approved analytical procedures are proposed in this submission. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. No changes to the approved analytical procedures are proposed in this submission. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

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Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

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Control of Excipients (3.2.P.4)

For the commercial formulation F-146/A, Methocel E5 Premium LV (Hypromellose, release-controlling polymer) is used.

The excipient control strategy was developed in accordance with USP-NF and FDA Guidance for Industry: Immediate Release Solid Oral Dosage Forms.

Selection Justification

Methocel E5 Premium LV was selected as the commercial grade of hypromellose on the basis of improved blend uniformity in geometric dilution studies.

Commercial Control Strategy — Excipients

Component	Function	Grade per current specification
Hypromellose	release-controlling polymer	Methocel E5 Premium LV
Reference		SPEC-EX-6477 Rev 06

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Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data.

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Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

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Process Validation Summary

Process performance qualification was executed on three consecutive commercial-scale batches. All predefined acceptance criteria for critical process parameters and critical quality attributes were met, supporting a conclusion of a validated state of control.

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